

Video Analytics Technology: Turning CCTV Cameras Into Proactive Tools

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RICHA CHAKRAVARTY

With technology advancing at a rapid pace, it is inevitable that there will be a continual stream of new products and solutions in the evolving security industry. Video analytics is the latest solution to generate immense interest within the industry. Analysts like IMS predict that the market for analytics software alone will be worth hundreds of millions of dollars within the next two to three years. The rationale be-

hind the strong acceptance of this solution is its capability to reduce human interventions to a great extent.

What is video analytics technology?

Video analytics or intelligent video surveillance (IVS) is a technology that uses software to automatically identify specific objects, behaviour or attitudes in video footage. It is basically used for intrusion detection. It transfers to the software the human interventions normally required to

detect intrusions. It has the capability to analyse a video frame or a group of video frames to detect, classify, recognise or identify a predetermined event that has been programmed through the IVS software. It uses mathematical algorithms to detect moving objects in an image and filter non-relevant movements. It then creates a database that records the attributes of all the objects detected and their movement.

Video analytics technology works best for large infrastructures, such as a mass transit system, where over a thousand surveillance cameras may be deployed. These installations represent a huge amount of video to transmit, view and archive, making it impossible for a human monitor to analyse all these video recordings in order to detect suspicious behaviour or events.

Explains Ramakrishna Sundaram, executive vice president and head, Building Technologies Division, Infrastructure and Cities Sector, Siemens, "When a large number of cameras cover a wider area, it becomes difficult to depend on the human eye alone. Here we need to depend upon IVS software, for example, to accurately recognise a face in a crowd or the number plate of a moving car."

Video analytics systems may be set up for a centralised or distributed architecture. In centralised architecture, all intelligent processing is concentrated in the same place, whereas



Video analytics intelligently detects suspicious movement